

# Chapter 9: The Operations Pillar in Practice — Revenue Cycle, HCC Coding, Credentialing, and Workforce in a Value-Based Environment

*The Operational Infrastructure That Makes or Breaks VBC Contracts: Clinical Documentation Excellence, Risk Adjustment Strategy, Credentialing Efficiency, and the Workforce Operations Framework for Transformed Health Systems*

## Operations Where Money Actually Lives

Healthcare transformation is ultimately executed through the operational infrastructure that most health system leaders think of as administrative — revenue cycle management, clinical documentation, credentialing, workforce management. These functions are not adjacent to clinical transformation; they are the substrate on which it runs. An organization that redesigns its care model for value-based care but fails to adapt its revenue cycle, clinical documentation, and workforce operations will achieve neither the quality outcomes nor the financial results the new model promises.

This chapter develops the operational mechanics of healthcare transformation — the specific processes, metrics, and management disciplines that determine whether value-based care contracts produce the results they are designed for. It is organized around four operational domains: revenue cycle transformation for VBC, HCC coding and risk adjustment excellence, credentialing efficiency, and the workforce operations framework.

## Section 1: Revenue Cycle Transformation — From FFS Optimization to VBC Management

Revenue cycle management under fee-for-service is a known discipline: charge capture, coding accuracy, claim submission, denial management, underpayment identification. Hospitals and physician practices have invested heavily in FFS RCM over the past two decades, and the industry has developed sophisticated tools and processes for maximizing FFS revenue. None of this investment is wasted under VBC — but all of it must be extended.

VBC revenue cycle adds three functions that FFS RCM does not require: attribution management (ensuring that patients are correctly attributed to the organization for shared savings calculation), total cost of care monitoring (tracking what attributed patients cost across all providers and settings, not just at your organization), and VBC contract reconciliation (understanding and disputing the financial calculations that determine shared savings or loss settlements).

### Attribution Management — The VBC-Specific Revenue Function

Under FFS, revenue is determined by what care the organization delivers. Under VBC, revenue is determined partly by which patients are attributed to the organization — and attribution errors can silently reduce shared savings or inflate shared losses by millions of dollars. Every organization in a significant VBC arrangement needs an attribution management function.

| Attribution error type | How it happens                          | Financial impact                         | Remediation                                        |
|------------------------|-----------------------------------------|------------------------------------------|----------------------------------------------------|
| False attributions     | Patients who receive care elsewhere are | Your organization bears cost of care you | Regular attribution list review; proactive patient |

|                                 |                                                                                                                                   |                                                                                                           |                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | attributed to your organization due to a primary care visit that triggered attribution before the patient transferred care        | did not deliver and cannot manage                                                                         | outreach to confirm active care relationship; request attribution corrections from payer using patient-level evidence                                                                         |
| Missing attributions            | Your highest-cost, highest-need patients are not attributed because they lack a primary care visit that would trigger attribution | You lose the opportunity to earn shared savings for care management programs that benefit this population | Proactive outreach to unattributed high-risk patients; care management programs that include primary care visits to establish attribution; PCMH recognition to improve attribution algorithms |
| Incorrect benchmark attribution | Attribution list errors inflate or deflate the risk adjustment factor applied to your benchmark                                   | Incorrect RAF produces a benchmark that doesn't reflect your population's actual cost burden              | Compare attribution list to actual patient panel; use clinical data to validate risk adjustment; submit corrected diagnoses codes through formal channels                                     |
| Retroactive attribution changes | Payer changes attribution methodology mid-contract, shifting patients in or out of your attributed population                     | Shared savings calculations change unexpectedly after performance year ends                               | Negotiate prospective attribution methodology in contract; require prior notice of methodology changes; maintain your own attribution model to detect changes                                 |

Figure 9.1 — Attribution error taxonomy and remediation. Source: HTR Operations pillar framework; ACO REACH and MSSP operational guidance.

**Total Cost of Care Monitoring — Seeing What You Cannot Bill**

The most difficult revenue cycle adaptation for FFS-trained teams is monitoring costs they do not receive. Under VBC, the organization's financial performance is determined by the total cost of care for its attributed population — including care delivered by specialists, hospitals, skilled nursing facilities, home health agencies, and any other provider outside the organization. A primary care group in an ACO may be held accountable for whether their attributed patients receive unnecessary specialist imaging ordered by a cardiologist who has no financial relationship with the ACO.

Vermont's VHCURES all-payer claims database is the analytical foundation for total cost of care monitoring — providing claims data across all payers and provider types for attributed Vermont beneficiaries. The GMCB analytics vendor procurement, and individual hospital analytics investments through the RHT Program, are building the capability to use VHCURES data for real-time population health management rather than retrospective reporting.

**Section 2: HCC Coding Excellence — The Risk Adjustment Foundation**

Hierarchical Condition Category (HCC) coding is the mechanism through which Medicare risk adjustment translates patient diagnoses into the Risk Adjustment Factor (RAF) that scales a beneficiary's expected cost. In VBC arrangements, accurate HCC coding is not just a compliance requirement — it is the foundation of financial sustainability. An organization that systematically under-codes its patients' diagnoses will have a benchmark set too low for its population's actual cost burden, and will appear to be underperforming even when its care management is excellent.

## The HCC Model — How It Works

Medicare's HCC model assigns each patient a RAF based on their demographic characteristics (age, sex, Medicaid status, disability) and their active chronic conditions as documented in claims. The RAF multiplies the base capitation or benchmark rate — a patient with a RAF of 1.5 has a benchmark 50% higher than the average Medicare beneficiary. Accurate diagnosis coding is essential because the RAF is calculated from diagnosis codes on claims; diagnoses that are not coded do not affect the RAF.

Vermont's rural hospital population has two characteristics that make HCC coding excellence especially important: high prevalence of complex chronic conditions (diabetes, COPD, CHF, CKD) that generate large HCC score differences based on coding specificity, and significant dually eligible (Medicare-Medicaid) population that has additional risk adjustment mechanisms. Vermont hospitals that do not code to the highest level of specificity are systematically under-representing their population's complexity and receiving benchmarks that do not reflect actual cost burden.

## The HCC Gap Closure Process

The HCC gap closure process — identifying diagnoses that should be coded based on clinical evidence but are not currently documented in claims — is one of the highest-ROI operational investments available to organizations in VBC arrangements. The process has four steps:

- **RAF analysis: Compare the organization's average RAF against national and regional benchmarks for comparable populations. A RAF significantly below benchmark for a population with high chronic disease prevalence signals a coding gap.**
- **Retrospective chart review: Review clinical documentation for attributed patients to identify diagnoses that are clinically evident (in problem lists, clinical notes, medication lists) but not documented in claims codes. Each uncoded HCC condition represents a missed RAF point.**
- **Prospective coding education: Educate clinical staff on HCC coding specificity requirements — the difference between coding 'diabetes' (HCC 19, lower RAF) versus 'diabetes with diabetic CKD' (HCC 18 + HCC 136, higher RAF) when clinical evidence supports the more specific code.**
- **Annual wellness visits: AWWs are the primary vehicle for capturing annual HCC updates — CMS requires that chronic conditions be documented at least annually to be included in RAF calculations. Systematic AWW completion for the attributed population is both a quality metric and an HCC gap closure mechanism.**

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## Vermont in Practice: HCC Coding at Vermont Community Hospitals

Oliver Wyman's Act 167 assessment of Vermont's hospital system did not specifically examine HCC coding accuracy, but the management and administrative cost data tells a related story: Vermont PPS hospitals spend \$2,730 per adjusted discharge on management and administrative costs against a benchmark of \$1,427. Part of this gap reflects genuine administrative inefficiency — systems and processes that could be streamlined. But part of it also reflects underinvestment in the clinical documentation and coding infrastructure that makes revenue cycle management accurate under both FFS and VBC.

Vermont hospitals entering the AHEAD global budget model in 2027 will have their budgets set based on historical Medicare spending adjusted for their population's risk complexity. An organization that has under-coded its population's HCC conditions will receive a lower global budget than its actual cost burden warrants — and will need to manage within that lower budget for the full performance year before the error can be corrected. Getting HCC coding right before AHEAD launches is not optional — it is foundational financial preparation.

Recommended action: Every Vermont hospital should conduct a retrospective HCC gap analysis for their AHEAD-attributed Medicare population before January 2027. The analysis should compare clinical documentation to coded diagnoses, identify the largest gaps by HCC category, and develop a provider education and AWW completion program to close the gaps prospectively.

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## Section 3: Credentialing Efficiency — The Hidden Operations Bottleneck

Provider credentialing — the process of verifying a clinician's education, training, licenses, and competency before granting hospital privileges or health plan participation — is one of the most operationally consequential and consistently underinvested functions in American healthcare administration. The average credentialing cycle takes 90-120 days for a new provider. In a state with a 370 FTE primary care shortage by 2030, a 90-day credentialing delay represents 90 days of foregone care — and foregone revenue — for every provider recruited.

### Credentialing as a Transformation Bottleneck

Vermont's transformation agenda accelerates the credentialing importance for two specific reasons. First, regionalization — moving to a tiered hospital network where patients follow clinical protocols to Centers of Excellence — requires providers to have privileges at multiple facilities, not just their home institution. A surgeon designated as a COE orthopedic provider must be credentialed at the regional COE facility to practice there, and the credentialing process must occur before the regionalization takes effect. Vermont hospitals that do not begin cross-facility credentialing before COE designations are finalized will face delays in implementing the regionalization blueprint.

Second, scope-of-practice expansion — enabling APRNs, PAs, and other non-physician clinicians to practice at the top of their licensure — requires updated credentialing standards and privilege delineations at every facility. Oliver Wyman's workforce recommendation to expand professional licensure and practice scope is not a paperwork exercise; it requires systematic credentialing policy updates at 14 hospitals, coordinated through the Vermont CIN's shared services infrastructure.

### The Credentialing Efficiency Framework

Three operational improvements reduce credentialing cycle time from 90-120 days to 45-60 days without reducing verification rigor:

- **Primary source verification automation: Replace manual verification of medical school diplomas, residency completions, and license verifications with automated primary source connections to NPDB, state licensing boards, and medical schools. Several credentialing platforms (VerityStream, Symplr, symplr MD-Staff) provide automated PSV that reduces verification time from weeks to hours.**
- **Continuous monitoring replacing point-in-time verification: Implement ongoing license and sanction monitoring that alerts the credentialing office when a provider's license status changes, rather than discovering changes at 2-year re-credentialing intervals. Continuous monitoring reduces risk while eliminating the administrative burden of complete re-verification at each cycle.**
- **Unified credentialing for the Vermont CIN: Vermont's 14 hospitals sharing a common credentialing database — so that a provider credentialed at one CIN member facility has their verified credentials available to all member facilities — reduces the per-facility credentialing burden for providers practicing at multiple sites. This is a direct shared service application that Oliver Wyman's CIN design makes possible.**

## Section 4: Workforce Operations — Managing the Care Team in a Transformed System

The Operations pillar's workforce domain encompasses three distinct management challenges that often get conflated: workforce supply (recruiting enough providers), workforce deployment (organizing providers in care team models that maximize effective capacity), and workforce retention (reducing attrition that compounds the supply problem). Vermont faces challenges in all three domains, and the solutions are different.

## The Workforce Supply Challenge — What Can Actually Be Done

Vermont projects a 370 FTE primary care shortage by 2030. Physician recruitment pipelines take 7-10 years from medical school enrollment to practice — making it impossible to recruit Vermont out of a 2030 shortage with current policies. The supply interventions that can produce meaningful results within the relevant timeframe are:

- Scope-of-practice expansion for APRNs and PAs: Oliver Wyman specifically recommended this. Vermont APRNs operating under full practice authority (without physician supervision requirements) can see three patients per hour in PCMH settings, effectively adding primary care capacity equivalent to additional physicians without waiting for new physicians to train.
- International Medical Graduate (IMG) recruitment and J-1 visa waiver programs: Vermont has J-1 visa waiver slots through the Conrad 30 program, enabling internationally trained physicians to practice in underserved areas in exchange for 3-year service commitments. Active management of this program — filling all available slots, targeting Northeast Kingdom communities — can meaningfully supplement domestic recruitment.
- Loan repayment and service obligation programs: Vermont's RHT Program workforce investments include tuition assistance and loan repayment with 5-year service obligations. The evidence on loan repayment is consistent: it works for early-career providers with high debt burdens. Vermont's program should be structured with the largest awards for the communities with the greatest access gaps.
- Regional medical education investment: The evidence on training location and practice location is strong — physicians trained in rural settings are more likely to practice in rural settings. Vermont's partnership with University of Vermont College of Medicine, and potential expansion of clinical training sites in Northeast Kingdom communities, is a 7-10 year investment with compound returns.

## Workforce Deployment — Team-Based Care at Scale

Oliver Wyman's finding that Vermont does not have an absolute primary care shortage but may have a care delivery model problem is operationally actionable. A primary care team in which physicians see 3 patients per hour — supported by medical assistants, care coordinators, CHTs, and behavioral health care managers who handle pre-visit preparation, care gap follow-up, and care coordination — has twice the effective capacity of a physician practicing in a traditional solo model.

Vermont's Blueprint PCMH model embeds this team-based care logic in its recognition standards. The operational challenge is that many Vermont practices — particularly smaller rural practices — lack the staffing depth to implement true team-based care. A rural practice with one physician and one MA cannot implement a full care team model regardless of how well the physician is trained. The RHT workforce investments, specifically the APRN and CHW training programs, are designed to build the care team depth that team-based care requires.

## Workforce Retention — The Math of Turnover

Physician and clinician turnover in rural healthcare is both a symptom of system stress and a cause of it. Every physician departure costs approximately \$500,000-\$1 million to replace (recruitment costs, onboarding time, lost productivity, temporary coverage costs). In a system where 14 hospitals are competing for a limited rural provider pool, high turnover is financially devastating — and the signal it sends to communities about healthcare stability intensifies the system fragility it creates.

The operational drivers of rural provider retention are well-documented: burnout driven by documentation burden (addressed by AI scribe technology), isolation from professional peers (addressed by telemedicine networks and CIN peer connectivity), inadequate support staff (addressed by team-based care investment), housing unavailability (addressed by Oliver Wyman's first imperative), and work-life balance limitations in single-provider practices (addressed by coverage sharing agreements within the CIN network). Vermont's transformation agenda addresses all of these — the question is whether the investments are made before the next wave of provider departures.

Section 5: Administrative Simplification — The Vermont Agenda

Oliver Wyman's community engagement found that providers across Vermont cited administrative complexity — prior authorization, CON processes, billing burden, duplicative reporting requirements — as major constraints on care delivery efficiency. Administrative simplification is not a soft nicety; it is an Operations pillar requirement that directly affects the effective capacity of Vermont's clinical workforce.

The Vermont Administrative Simplification Priorities

AHS and GMCB have identified three administrative simplification priorities that must be addressed in the Statewide Strategic Plan:

- **Prior authorization reform:** As developed in Chapter 5, Vermont's Act 167 and Act 68 both address prior authorization complexity. The specific operational target is moving to gold-carding arrangements — automatic approval for providers with demonstrated compliance histories — and eliminating PA requirements for evidence-based, guideline-concordant care in high-volume service categories.
- **CON process streamlining:** Vermont's Certificate of Need process for hospital capital investments was cited as a barrier to efficient regionalization — providers cannot build the facilities and purchase the equipment that transformation requires if CON approval takes 12-18 months and involves litigation risk. Act 167 directed GMCB to evaluate CON process improvements; Vermont's transformation requires faster turnaround for transformation-aligned capital investments.
- **Standardized quality reporting:** Vermont's quality reporting landscape requires providers to submit data to multiple programs (Blueprint, GMCB, AHEAD, commercial payers) with overlapping but non-identical measures. The Blueprint Quality Measure Crosswalk begins to address this; the Statewide Strategic Plan should commit to further standardization that eliminates duplicative data submission for commonly required measures.

Key Concepts

| Term                                  | Definition                                                                                                                                                                                                                                                                |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attribution management                | The operational function of ensuring that patients are correctly attributed to an ACO or VBC organization for shared savings calculations, including regular attribution list review, correction request processes, and proactive outreach to confirm care relationships. |
| HCC (Hierarchical Condition Category) | The Medicare risk adjustment model that assigns RAF scores based on patient diagnoses. Accurate and complete HCC coding is essential for VBC contract financial management — under-coded diagnoses produce benchmarks that understate the population's expected cost.     |
| RAF (Risk Adjustment Factor)          | The multiplier applied to a benchmark or capitation rate based on a patient's HCC profile. A RAF of 1.5 means the patient is expected to cost 50% more than the average Medicare beneficiary.                                                                             |
| AWV (Annual Wellness Visit)           | A Medicare-covered annual visit focused on health risk assessment, preventive care planning, and chronic condition documentation. The primary mechanism for capturing annual HCC updates; systematic AWV completion is both a quality metric and an HCC gap closure tool. |

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|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HCC gap closure                   | The process of identifying diagnoses that are clinically evident but not currently documented in claims, through retrospective chart review and prospective coding education. A high-ROI operational investment for organizations entering VBC arrangements.                                |
| Gold carding                      | A prior authorization arrangement in which providers with demonstrated compliance histories receive automatic approval without requiring case-by-case review. Vermont's administrative simplification agenda includes gold carding as a priority PA reform.                                 |
| Primary source verification (PSV) | The credentialing process of directly confirming provider credentials with the issuing source (medical school, residency program, state licensing board) rather than accepting secondary documentation. Automation of PSV reduces credentialing cycle time from weeks to hours.             |
| Conrad 30 program                 | A federal J-1 visa waiver program enabling internationally trained physicians to practice in medically underserved areas in exchange for 3-year service commitments. Vermont has 30 annual waiver slots; active management of this program is a near-term primary care supply intervention. |

Sources: HTR Operations pillar framework; Oliver Wyman Act 167 Report (2024); Act 68 of 2025; Vermont RHT Program Application (November 2025); CMS ACO REACH and MSSP operational guidance; CMS Medicare Risk Adjustment documentation; AHS Transformation Reports (2025); Vermont Blueprint for Health Annual Report (2024).